

# TECHNO COMMERCIAL COMPARISON

19.6 MW DC Solar Plant -

Prepared for: Client - N/A

Date: July 2026

|                                                                      |                                                                      |                                                                       |                                                                    |                                                           |
|----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Redren - Equity IRR</b><br><b>38.94%</b><br><i>NPV Rs.42.9 Cr</i> | <b>Waaree - Equity IRR</b><br><b>36.80%</b><br><i>NPV Rs.43.2 Cr</i> | <b>Module3 - Equity IRR</b><br><b>36.80%</b><br><i>NPV Rs.43.2 Cr</i> | <b>Payback Period</b><br><b>3/3/3 yr</b><br><i>equity recovery</i> | <b>Recommended</b><br><b>Redren</b><br><i>Score: 90.0</i> |
|----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|

## Project & Financial Highlights

|                     |               |               |               |
|---------------------|---------------|---------------|---------------|
| Project DC Capacity | 19.6 MW DC    | 19.6 MW DC    | 19.6 MW DC    |
| Module Brand        | Redren        | Waaree        | Module3       |
| Module Wattage      | N/A Wp        | N/A Wp        | N/A Wp        |
| Modules Required    | 32,667        | 31,870        | 31,870        |
| Total Project Cost  | Rs. 62.72 Cr  | Rs. 66.64 Cr  | Rs. 66.64 Cr  |
| Equity @ 30%        | Rs. 18.82 Cr  | Rs. 19.99 Cr  | Rs. 19.99 Cr  |
| Year 1 Generation   | 32,623 MWh    | 33,137 MWh    | 33,137 MWh    |
| Equity IRR          | 38.94%        | 36.80%        | 36.80%        |
| NPV @ 10% Disc.     | Rs. 42.92 Cr  | Rs. 43.23 Cr  | Rs. 43.23 Cr  |
| LCOE                | Rs. 0.838/kWh | Rs. 0.852/kWh | Rs. 0.852/kWh |
| Payback Period      | 3 years       | 3 years       | 3 years       |

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| <b>RECOMMENDATION: Redren - Equity IRR 38.94%   NPV Rs. 42.92 Cr   LCOE Rs. 0.838/kWh   Equity Rs. 18.82 Cr</b> |
|-----------------------------------------------------------------------------------------------------------------|

Disclaimer: Based on manufacturer datasheets and standard financial modelling. Actual returns may vary based on site conditions, financing terms, and tariff rates.

## 1. Executive Summary

This report presents a technical and financial comparison of 3 DCR-compliant solar PV modules for a 19.6 MW DC ground-mount plant at the site. The following modules are evaluated: Redren, Waaree, Module3 on a Fixed Tilt configuration.

### Key Findings

- Redren: Equity IRR 38.94% | NPV Rs. 42.92 Cr | LCOE Rs. 0.838/kWh
- Equity: Rs. 18.82 Cr | Payback: 3 years
- Waaree: Equity IRR 36.80% | NPV Rs. 43.23 Cr | LCOE Rs. 0.852/kWh
- Equity: Rs. 19.99 Cr | Payback: 3 years
- Module3: Equity IRR 36.80% | NPV Rs. 43.23 Cr | LCOE Rs. 0.852/kWh
- Equity: Rs. 19.99 Cr | Payback: 3 years
- Waaree generates 4.4% more lifetime energy than Redren

### Recommendation

**Based on comprehensive multi-criteria analysis, Redren is RECOMMENDED (Weighted Score: 90.0/100). The final decision should align with the investor's return expectations and risk appetite.**

## 2. Project Details

| Project Parameter   | Redren                       | Waaree                       | Module3                      |
|---------------------|------------------------------|------------------------------|------------------------------|
| Customer Name       |                              |                              |                              |
| Company             |                              |                              |                              |
| Project DC Capacity | 19.6 MW DC                   | 19.6 MW DC                   | 19.6 MW DC                   |
| Location            |                              |                              |                              |
| Module Brand (1)    | Redren                       |                              |                              |
| Module Wattage (1)  | N/A Wp                       |                              |                              |
| Module Count (1)    | 32,667 nos                   |                              |                              |
| Land Area (1)       | 62.6 acres (25.3 ha)         |                              |                              |
| Module Brand (2)    | Waaree                       |                              |                              |
| Module Wattage (2)  | N/A Wp                       |                              |                              |
| Module Count (2)    | 31,870 nos                   |                              |                              |
| Land Area (2)       | 61.0 acres (24.7 ha)         |                              |                              |
| Module Brand (3)    | Module3                      |                              |                              |
| Module Wattage (3)  | N/A Wp                       |                              |                              |
| Module Count (3)    | 31,870 nos                   |                              |                              |
| Land Area (3)       | 61.0 acres (24.7 ha)         |                              |                              |
| Mounting Structure  | Fixed Tilt                   | Fixed Tilt                   | Fixed Tilt                   |
| Analysis Basis      | Frontside only (no bifacial) | Frontside only (no bifacial) | Frontside only (no bifacial) |
| Plant Life          | 25 years                     | 25 years                     | 25 years                     |

Land area uses a land-to-module-area factor of 3.0x for Fixed Tilt mounting. Actual requirement may vary based on row spacing, access roads, and site boundary.

## 3. Project Background

The 19.6 MW DC plant is at the site (Lat: N/A, Lon: N/A). Configuration: Fixed Tilt. DCR-compliant (indigenously manufactured modules required).

### Site-Specific CUF Assumptions

- Redren: 19.0%
- Waaree: 19.3%
- Module3: 19.3%

## 4. Module Technical Specifications

Specifications extracted directly from manufacturer datasheets.

| Parameter          | Redren             | Waaree        | Module3       | Unit   |
|--------------------|--------------------|---------------|---------------|--------|
| Technology         | Mono PERC (p-type) | N-type TOPCon | N-type TOPCon |        |
| Efficiency         | 21.30              | 22.77         | 22.77         | %      |
| Temp Coeff Pmax    | -0.350             | -0.300        | -0.300        | %/°C   |
| Degradation Y1     | 2.00               | 1.00          | 1.00          | %      |
| Degradation Annual | 0.55               | 0.40          | 0.40          | %      |
| Power Warranty     | 27                 | 30            | 30            | years  |
| Price              | 20                 | 22            | 22            | Rs./Wp |

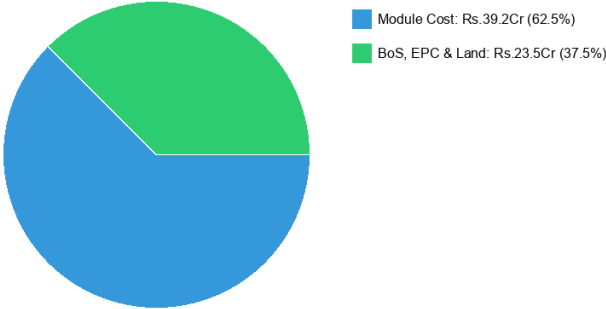
5. Financial Analysis & Projections

5.1 Capital Expenditure (CAPEX)

Total project cost: Rs. 62.72 Cr (Redren) vs Rs. 66.64 Cr (Waaree) vs Rs. 66.64 Cr (Module3).

| Cost Component (Rs. Cr) | Redren | Waaree | Module3 |
|-------------------------|--------|--------|---------|
| Module Cost             | 39.20  | 43.12  | 43.12   |
| BoS, EPC & Land         | 23.52  | 23.52  | 23.52   |
| Total Project Cost      | 62.72  | 66.64  | 66.64   |
| Equity @ 30%            | 18.82  | 19.99  | 19.99   |

Project Cost Breakdown (Total Rs. 62.7Cr)

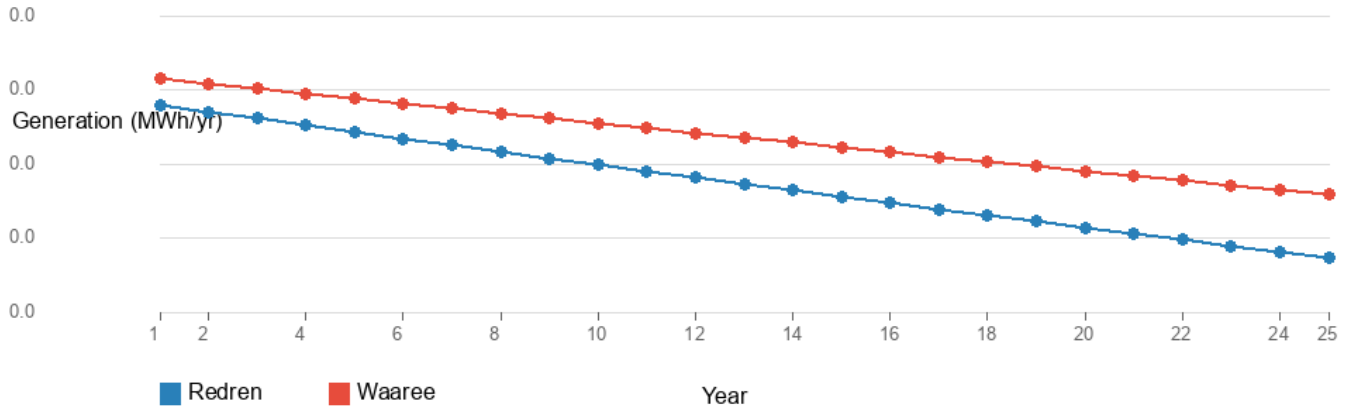


Cost breakdown: module cost typically represents 55-65% of total CAPEX.

5.2 Energy Generation & Revenue

Year 1: 32,623 MWh (Redren) | Year 1: 33,137 MWh (Waaree) | Year 1: 33,137 MWh (Module3). GHI N/A kWh/m2/yr, POA N/A kWh/m2/yr, PR N/A.

## Annual Energy Generation (MWh) - 25 Year Horizon

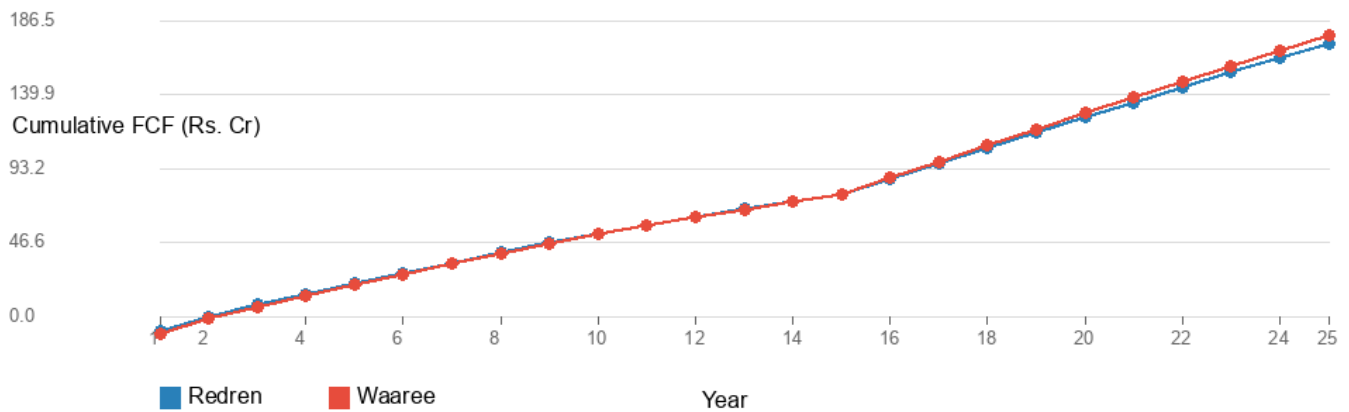


Annual generation over 25 years. Decline driven by module degradation; lower degradation rates sustain yield.

### 5.3 Cash Flow Analysis & Returns

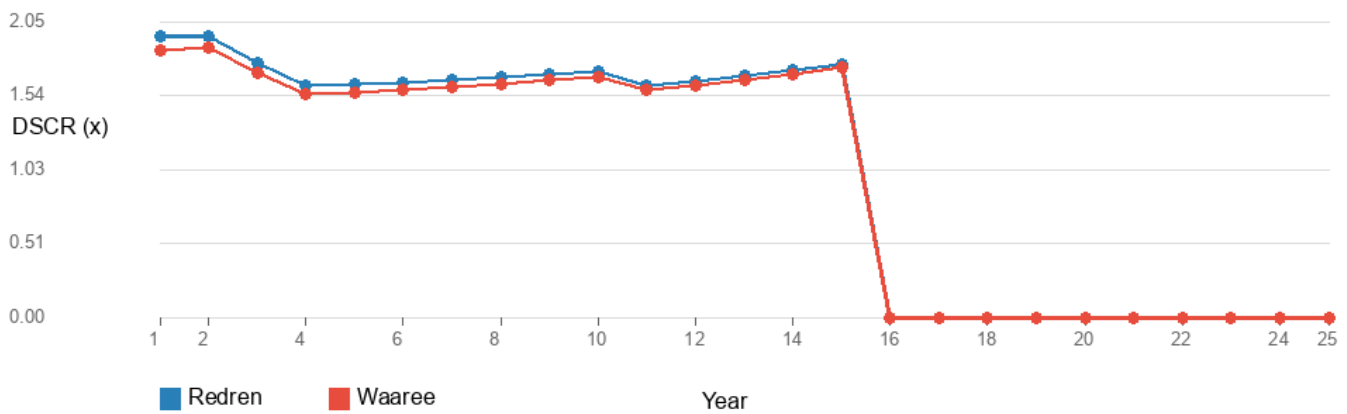
Model includes revenue, O&M (3% escalation), insurance, debt service (15-yr @ 9%), WDV depreciation, tax 25.17%.

## Cumulative Free Cash Flow to Equity (Rs. Crores)



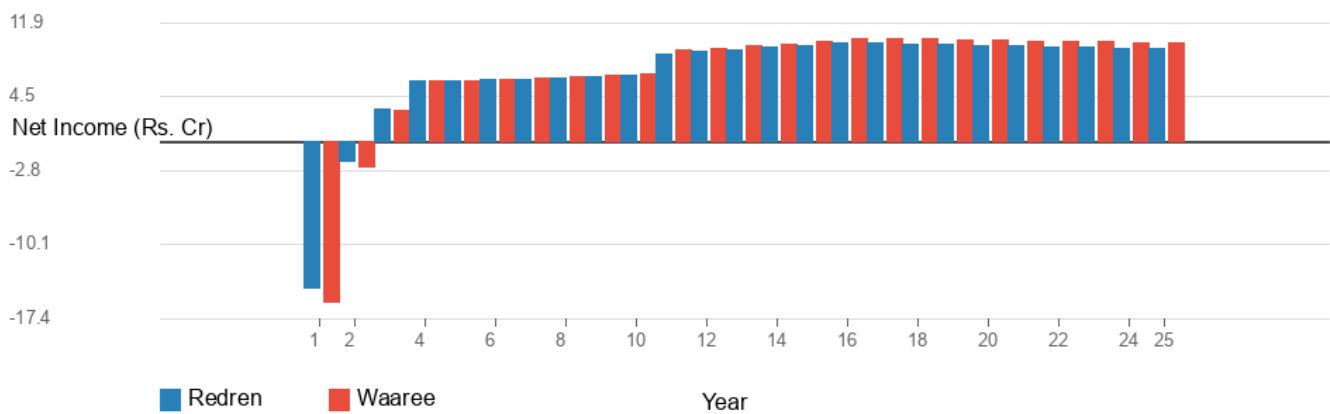
Cumulative FCF - payback reflects equity recovery timeline. Post-payback slope reflects ongoing returns.

### Debt Service Coverage Ratio (Debt Tenure Period)



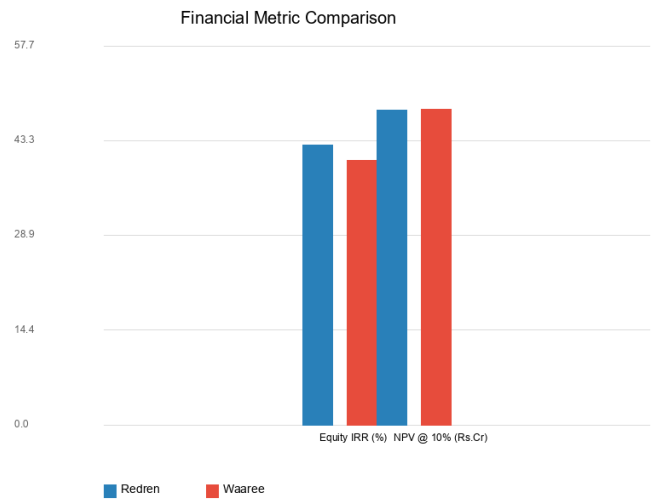
DSCR > 1.5x throughout the loan tenure confirms strong debt repayment capacity and bankability.

### Annual Net Income After Tax (Rs. Crores)



Net income after tax. Early years lower due to interest and WDV depreciation; improves as debt reduces.

6. Key Financial Metrics Comparison



IRR = annualised return on equity over 25 years. NPV = present value of all cash flows at WACC. Higher is better for both.

| Financial Metric            | Redren | Waaree | Module3 |
|-----------------------------|--------|--------|---------|
| Total Project Cost (Rs. Cr) | 62.72  | 66.64  | 66.64   |
| Equity Required (Rs. Cr)    | 18.82  | 19.99  | 19.99   |
| Annual Gen Y1 (MWh)         | 32,623 | 33,137 | 33,137  |
| Total Gen 25yr (GWh)        | 748.7  | 782.0  | 782.0   |
| CUF (%)                     | 19.0   | 19.3   | 19.3    |
| Revenue Y1 (Rs. Cr)         | 14.39  | 14.76  | 14.76   |
| Equity IRR (%)              | 38.94  | 36.80  | 36.80   |
| NPV @ 10% (Rs. Cr)          | 42.92  | 43.23  | 43.23   |
| LCOE (Rs./kWh)              | 0.838  | 0.852  | 0.852   |
| Payback Period (years)      | 3      | 3      | 3       |

6A. Multi-Criteria Scoring

Weighted multi-criteria scoring across 7 dimensions. Each criterion is min-max normalized (0-100). Higher total = better investment.

7. PVSyst-Style Loss Analysis

Detailed energy loss breakdown modeled on PVSyst methodology. Losses are applied sequentially from POA irradiance to grid injection.

8. Risk Analysis & Sensitivity

| Risk Factor        | Assessment                                                                               |
|--------------------|------------------------------------------------------------------------------------------|
| PPA Tariff Risk    | Rs. 0.50/kWh reduction reduces IRR by ~3-4%. All modules equally exposed.                |
| Generation Risk    | 10% lower CUF reduces IRR by ~4-5%. Lower temp coeff modules offer marginal protection.  |
| Interest Rate Risk | 1% rate increase reduces IRR by ~1.5%. Lower-debt module has slightly better resilience. |
| Degradation Risk   | Higher degradation impacts long-term returns. Lower annual degradation is preferred.     |
| Technology Risk    | PERC is mature; N-TOPCon and HJT are next-generation with longer-term upgrade potential. |
| DCR Compliance     | All modules are DCR-compliant, mitigating import and policy risks.                       |

8.1 Sensitivity Analysis

Module price is the single largest return lever (~3% IRR per Rs. 1/Wp change). Redren's price advantage of Rs. 2.0/Wp is a key driver of return differentials.

9. Conclusion & Recommendation

9.1 Comparative Assessment

| Redren             | Waaree             | Module3            |
|--------------------|--------------------|--------------------|
| IRR: 38.94%        | IRR: 36.80%        | IRR: 36.80%        |
| NPV: Rs.42.92 Cr   | NPV: Rs.43.23 Cr   | NPV: Rs.43.23 Cr   |
| LCOE: Rs.0.838/kWh | LCOE: Rs.0.852/kWh | LCOE: Rs.0.852/kWh |
| Degr.: 0.55%/yr    | Degr.: 0.4%/yr     | Degr.: 0.4%/yr     |
| TC: -0.350%/C      | TC: -0.300%/C      | TC: -0.300%/C      |
| Warr.: 27yr        | Warr.: 30yr        | Warr.: 30yr        |

9.2 Final Recommendation

**RECOMMENDATION: Redren is recommended based on Score: 90.0/100 | Equity IRR 38.94%, equity Rs. 18.82 Cr, and LCOE Rs. 0.84/kWh. DCR certification ensures bankability and investor confidence.**

Note: If maximising total lifetime returns is the priority, Module3's higher NPV (Rs. 43.23 Cr vs Rs. 42.92 Cr) makes it a compelling alternative.

**BANKABILITY STATEMENT**

This project demonstrates strong financial metrics (Min. DSCR > 1.5x, IRR > 35%, payback within 3 years) meeting standard project finance lending criteria. Conservative assumptions provide additional comfort to lenders and equity investors.